

Analyzing and Forecasting Household Electricity Consumption Using Statistical, Machine Learning and Deep Learning Approaches

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Abstract—In the current age of technology advancement, the demand for electricity is increasing and electricity is not only for daily usage but also plays an essential role in economic and social operations, and understanding the factors affecting electricity consumption is an effective key in load forecasting. Particularly, temperature has long been considered an important forecasting variable, but many studies have shown that its influence can vary depending on the data frequency, especially with hourly load data, where endogenous factors such as lag, diurnal cycle and system inertia dominate. This study uses a dataset of more than 10,000 hourly observations to assess the influence of temperature on electricity consumption and compare it with time variables.

The method combines descriptive statistics, testing and ten forecasting models including Linear Regression, ARIMA, ARIMAX, SARIMAX, Random Forest, Decision Tree, LSTM, XGBoost and SARIMAX–LSTM hybrid model. The results show that temperature has a significant but not dominant influence over the endogenous properties of the time series. Additionally, both of deep learning and hybrid models perform better simulation capabilities of nonlinear relationships, while traditional models still play an important benchmarking role. Therefore, this study provides a comprehensive view of hourly electricity consumption dynamics and its driving factors, supporting load planning and power system operation.

I. INTRODUCTION

It is obvious to clarify that electricity plays an essential role in modern life and the socio-economic system. Additionally, by understanding the factors affecting electricity consumption is important for load planning, demand forecasting, and grid operation optimization. Among which of them, including weather and especially temperature, has long been considered one of the most important forecasting factors (Pardo et al[1], 2002; Fan & Hyndman[4], 2010).

However, the influence of temperature on electricity consumption is not always strong, especially in hourly load data. At this frequency, electricity consumption may be more influenced by load lag, daily life cycle and system inertia (Hong[5], 2010; Hippert et al[6], 2001). Therefore, the impact

of weather must be evaluated in parallel with the endogenous attributes of the time series.

According to this research, which aims to evaluate the influence of weather, especially temperature, on hourly electricity consumption, and compare this influence with time variables (lag, rolling, seasonality). The data includes more than 10,000 hourly observations, with many meteorological variables and time series. The research method combines: Descriptive Statistics, Inferential Statistics, Regression Analysis, ARIMA, ARIMAX, SARIMAX, Random Forest, Decision Tree, LSTM, XGBoost Regression, Hybrid SARIMAX and LSTM.

The main object of this research support to determine the influence of temperature on hourly electricity consumption and compare it with the endogenous impact of the time string variables. This study is arranged as outlines: considering related work in Section II then we detail the methodology in Section III, present the results in Section IV, and conclude with insights and limitations in Section V.

II. RELATED RESEARCH

The relationship between temperature and electricity consumption has been recorded in many international studies. Pardo, Meneu & Valor [1] found that temperature increases lead to the demand for electricity, mainly by using cooling systems. Similarly, Al-Hamadi & Solima[2] have demonstrated that temperature is a crucial predictor in load prediction. Some research suggests the non-linear relationship between temperature and power demand. Fan & Hyndman [4] indicated that load surged in approximately high temperatures, creating a non-linear structure, making linear regression ineffective. Others implemented LSTM, such as C. Cai et al.[3] confirmed that deep learning can simulate this relation better. Regarding hourly data, Hong[5] and Hippert et al.[6] indicated that lag variables are more important than temperature because of

the habit of consumption and the cycle of strong days. This makes temperature just play a supporting role in a short-term prediction.

III. MATERIAL AND METHOD

A. Dataset

The dataset used in this study focuses on a household power consumption, recorded continuously over a specific sequence of time with a total of 605,260 observations, following minutes. The dataset includes the active_power value, representing instantaneous electricity consumption with many other technical and environmental variables such as current, voltage, reactive power, apparent power, power factor, temperature, humidity, atmospheric pressure and thermal sensation. These attributes help to fully describe the operating conditions of electrical equipment as well as the impact of weather factors on electricity consumption.

The wide observation period, with data recorded at each continuous time point, allows for clear identification of consumption cycles on an hourly, daily and weekly basis. At the same time, the dataset reflects actual fluctuations in electricity usage behavior, including peak consumption times and low load periods, contributing to a deeper analysis of both short-term characteristics and long-term trends of household electricity loads.

B. Statistical model

Descriptive statistics is a collection of methods to describe and summarize data to understand the basic characteristics of each variable before proceeding to the deep statistics. The indices such as mean, standard deviation, min-max, median, and percentiles help to show the trend of the center and the level distribution of data. Additionally, distribution shape measures such as skewness and kurtosis allow to recognize whether the data has deviation or outliers. This process plays a vital role in assessing the reasonableness of data, detecting unusual data, and showing the right direction to proceed for the next model.

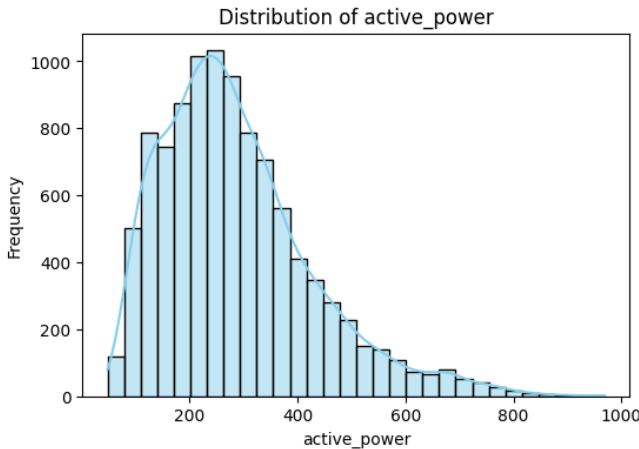


Fig. 1. Distribution and density estimation of household electricity consumption (active_power).

=== (Descriptive Statistics) ===											
	count	mean	std	min	Q1	median	Q3	max	CV(%)	Skewness	Kurtosis
active_power	10112.0	286.186	140.941	47.870	184.793	262.268	358.115	969.083	49.248	1.036	1.228
temp	10112.0	19.527	6.537	-4.000	14.809	19.476	24.289	38.065	33.480	-0.063	-0.339
feels_like	10112.0	18.715	6.648	-4.679	13.868	18.835	23.815	35.462	35.523	-0.177	-0.466
pressure	10112.0	1015.240	5.761	999.000	1011.000	1014.975	1019.150	1032.900	0.567	0.188	-0.391
speed	10112.0	2.625	1.947	0.000	1.090	2.265	3.873	10.290	74.180	0.821	0.066
weekday	10112.0	2.993	1.996	0.000	1.000	3.000	5.000	6.000	66.695	0.007	-1.244
is_weekend	10112.0	0.283	0.451	0.000	0.000	0.000	1.000	1.000	159.129	0.963	-1.073
ap_lag_1h	10112.0	286.157	140.934	47.870	184.793	262.268	357.994	969.083	49.251	1.037	1.229
ap_lag_24h	10112.0	286.442	141.120	47.870	184.926	262.418	358.835	969.083	49.266	1.034	1.215
ap_roll_24h	10112.0	286.239	105.373	96.555	219.527	257.582	325.279	668.514	36.813	1.323	1.605

Fig. 2. Descriptive statistical summary in the dataset.

Moreover, the study also applied statistical tests to assess the time and weather impact on the level of electricity consumption. Furthermore, the tests used include t-test, chi-square test, ANOVA and Pearson correlation coefficient. These tests provide additional information about the structure of the data and help determine the degree of dependence between similar variables

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

where values of r near 1 or -1 is shown as strong linear relationships. To further examine statistical differences between groups, t-tests and ANOVA were employed. The two-sample t-test was applied to compare weekday and weekend consumption levels, computed as:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

where $\bar{x}_1, \bar{x}_2$ are sample means, s_1, s_2 are sample standard deviations, and n_1, n_2 represent sample sizes. Chi-square testing was also used to detect distributional differences across categorical consumption groups, using:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i and E_i indicate observed and expected frequencies. These inferential tests help determine whether observed variations in consumption behavior are statistically meaningful.

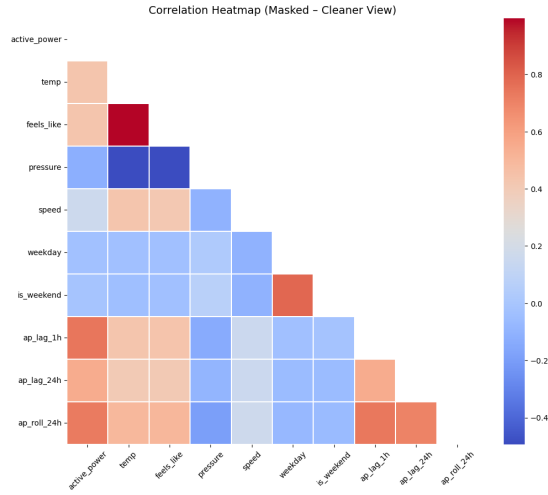


Fig. 3. Correlation heatmap of electricity consumption and weather-related variables.

The correlation heatmap above reveals that temperature-related variables have the strongest positive correlation with *active_power*, whereas variables such as humidity and pressure have comparatively weaker effects. Lagged features, including *ap_lag_1h* and *ap_lag_24h*, exhibit high correlation with the target variable due to the cyclic and autoregressive nature of electricity usage.

Finally, multiple linear regression was used to evaluate the linear influence of weather variables and lag-based features on electricity consumption. The regression model is formulated as:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \dots + \beta_k x_{k,t} + \varepsilon_t$$

where y_t represents electricity consumption at time t , and $x_{k,t}$ includes predictors such as temperature, humidity, and lagged consumption values. This model establishes a baseline for understanding linear relationships and serves as a comparison point for more advanced forecasting models introduced in subsequent sections.

C. Forecasting Models

About the forecasting time series models is applied to capture the endogenous structure of hourly electricity consumption and evaluate exogenous impact components such as temperature, humidity and wind speed. Therefore, this study focuses on three classical forecasting models including ARIMA, ARIMAX, and SARIMAX. These models serve as the foundation for understanding baseline temporal dynamics before applying machine learning approaches.

1) ARIMA: is Autoregressive Integrated Moving Average (ARIMA) model is a fundamental linear time-series forecasting approach used to capture the endogenous structure of a univariate series. ARIMA models rely solely on past values and past errors of the target variable, making them suitable for detecting autoregressive behavior, trend components, and short-term temporal dependencies. In electricity forecasting,

ARIMA serves as a baseline model to evaluate whether the load can be predicted effectively without using external factors such as temperature or weather conditions.

The ARIMA(p, d, q) model is defined as:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

Where:

- Y_t : target variable (*active_power*)
- c : constant term
- ϕ_i : autoregressive coefficients
- θ_j : moving-average coefficients
- ε_t : error term

The ARIMA model is particularly useful for identifying the effect of internal load dynamics, including hour-to-hour dependency, noise levels, and short-term fluctuations

2) ARIMAX: is Autoregressive Integrated Moving Average with Explanatory Variables (ARIMAX) extends the ARIMA model by incorporating exogenous predictors that may influence electricity consumption. This model functions as a multivariate regression combined with autoregressive (AR) and moving-average (MA) terms. ARIMAX is well-suited for capturing additional effects from external factors such as temperature, humidity, and lagged weather variables.

Unlike ARIMA, which is restricted to univariate patterns, ARIMAX allows the integration of additional explanatory information. These exogenous variables represent external influences that affect electricity demand but are not part of the target series itself. Typical examples include meteorological indicators (temperature, humidity), categorical time-related features (weekday, weekend), and Boolean events.

The ARIMAX(p, d, q) model is expressed as:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \beta X_t + \varepsilon_t$$

Where:

- Y_t : target variable (*active_power*)
- X_t : vector of exogenous variables (e.g., temp, feels_like)
- β : coefficient corresponding to each exogenous factor
- ϕ_i : autoregressive parameters
- θ_j : moving-average parameters
- ε_t : error term

ARIMAX is used in this study to determine whether weather variables significantly improve forecasting performance.

3) SARIMAX The Seasonal ARIMAX (SARIMAX) model enhances ARIMAX by incorporating seasonal components to account for repeating load patterns such as daily or weekly cycles. Electricity consumption typically exhibits strong seasonality, especially with hourly data showing clear 24-hour repetition cycles.

The SARIMAX model is written as:

$$SARIMAX(p, d, q)(P, D, Q)_s$$

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \beta X_t + \sum_{k=1}^P \Phi_k Y_{t-ks} + \sum_{l=1}^Q \Theta_l \varepsilon_{t-ls} + \varepsilon_t$$

Where:

- (P, D, Q) : seasonal AR, differencing, MA terms
- s : seasonal cycle length (24 for hourly consumption)
- Φ_k, Θ_l : seasonal AR and MA coefficients
- X_t : exogenous weather variables

SARIMAX is preferred for hourly load forecasting because it captures both external weather effects and strong cyclical patterns such as daytime peaks and nighttime reductions. This makes it one of the most suitable linear models for electricity load prediction.

D. Machine Learning and Deep Learning Models

Regarding to both of machine learning and deep learning models which are applied to capture nonlinear dependencies and long-term temporal patterns in hourly electricity consumption. These models complement classical forecasting approaches by learning complex relationships between consumption behavior, weather factors, and lagged features. The following subsections present a summary of the models used in this study.

1) Decision Tree Regression

The Decision Tree Regression model is a monitored learning method that partitions the feature space into regions with similar target values through recursive binary splitting. The model predicts electricity consumption by learning decision rules based on weather features, time indicators, and lagged consumption.

For every split, the model collects a attribute and threshold that minimize the impurity of the two resulting subsets. The impurity measure for regression is typically defined as:

$$G(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y}_S)^2$$

where S is a subset of the data and $\bar{y}_S$ is the mean target value within that subset.

Where:

- y_i : actual consumption value
- $\bar{y}_S$: average consumption in region S
- $G(S)$: impurity (variance) within node S

Decision Trees are fast and interpretable, making them suitable for identifying nonlinear relationships between electricity consumption and external variables. 2) Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and averages their predictions to reduce overfitting and variance. Each tree is trained on a bootstrap sample of the data, and at each node a random subset of features is considered for splitting, improving model robustness.

The Random Forest prediction is computed as:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B f_b(x)$$

Where:

- B : number of trees
- $f_b(x)$: prediction of the b -th tree

Random Forest is effective for modeling nonlinear effects of weather variables and lag-based features on electricity demand.

3) XGBoost Regression

XGBoost is a gradient boosting framework designed for high performance and scalability. The model constructs trees sequentially, where each new tree fits the residuals of the previous ensemble. XGBoost uses regularization to avoid overfitting, making it suitable for high-dimensional feature sets involving time, weather, and lagged variables.

The objective function is defined as:

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^t \Omega(f_k)$$

with regularization term:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

Where:

- l : loss function (e.g., squared error)
- T : number of leaves
- w_j : leaf weights
- γ, λ : regularization parameters

XGBoost is highly competitive for electricity forecasting due to its ability to model complex nonlinear structures efficiently.

4) LSTM

The Long Short-Term Memory (LSTM) network is a specialized recurrent neural network (RNN) designed to capture long-term temporal dependencies. It uses memory cells and gating mechanisms to control information flow, making it effective for modeling sequential patterns in hourly electricity consumption.

The LSTM cell is governed by the following equations:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

Where:

- f_t, i_t, o_t : forget, input, output gates
- C_t : cell state
- h_t : hidden state

LSTM is highly effective for observing long-term daily and weekly seasonal patterns in electricity usage.

5) Hybrid SARIMAX–LSTM

The hybrid SARIMAX–LSTM model combines the strengths of linear time-series modeling and deep learning. SARIMAX captures trend, seasonality, and linear temporal dependencies, while LSTM learns nonlinear residual patterns.

The hybrid structure decomposes the series as:

$$Y_t = L_t + N_t$$

where L_t is the linear component modeled by SARIMAX and N_t is the nonlinear component learned by LSTM.

Where:

- L_t : seasonal-linear component
- N_t : nonlinear residual component

This hybrid approach improves forecasting accuracy when neither linear nor nonlinear models alone sufficiently capture the dynamic load behavior.

IV. RESULTS

A. Statistical Results

The boxplot of *active_power* (Fig. 4) shows substantial variability in household electricity consumption, with a median around 262 W and numerous upper outliers exceeding 700–900 W. This confirms the presence of occasional high-demand periods, consistent with peak appliance usage. The statistical models also visualize the positive trends for temperature, feels-like temperature, and wind speed, while humidity and pressure exhibit weak negative associations.

Pearson’s correlation analysis provides quantitative confirmation of these observations. As shown in Table I, temperature ($r = 0.433$) and feels-like temperature ($r = 0.437$) have the strongest positive correlations with electricity consumption, followed by wind speed ($r = 0.161$). Humidity ($r = -0.139$) and pressure ($r = -0.124$) demonstrate weak negative correlations. All variables have p -values of 0.0, indicating statistically significant linear relationships.

TABLE I
PEARSON CORRELATION RESULTS

Variable	Correlation (r)	p-value
Temp	0.433	0.0000
Feels_like	0.437	0.0000
Humidity	-0.139	0.0000
Pressure	-0.124	0.0000
Wind Speed	0.161	0.0000

To evaluate temporal effects, the two-sample t-test comparing weekday and weekend consumption showed no statistically significant difference ($p = 0.2014$), as demonstrated in Fig. 5. This shows that household consumption patterns remain relatively stable between weekdays and weekends.

In contrast, the one-way ANOVA across seven days of the week yielded a statistically significant result ($F = 8.0923$, $p = 9.62 \times 10^{-9}$), confirming meaningful differences in daily consumption patterns (Fig. 6). This suggests that while overall weekday-weekend usage is similar, individual days exhibit distinct behavioral load characteristics.

These findings collectively demonstrate that electricity consumption is primarily influenced by temperature conditions, lagged usage patterns, and weekly behavioral cycles rather than daily calendar categories such as weekday versus weekend.

To further assess the extent to which these statistical relationships translate into predictive capability, a multiple linear regression baseline was evaluated. The regression model incorporates the same set of weather and temporal variables used in the statistical analysis. Consistent with the correlation trends, the model was able to capture medium-scale fluctuations in consumption; however, it struggled with nonlinear effects and rapid spikes. This limitation is reflected in its test-set performance, achieving an MAE of 60.55, an RMSE of 77.88, and a MAPE of 39.02%. These results confirm that while linear regression provides a useful interpretability baseline, more advanced forecasting models are required to effectively learn the complex behavior present in hourly electricity demand.

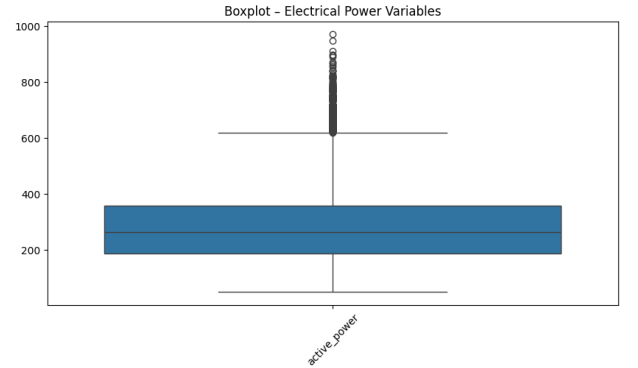


Fig. 4. Boxplot of active power.

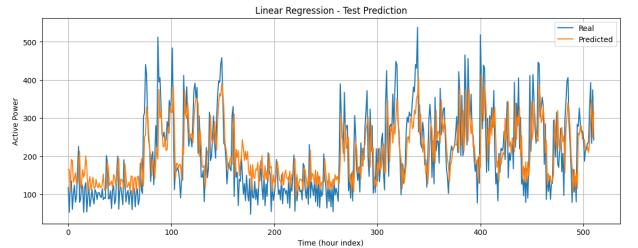


Fig. 7. Linear Regression test-set predictions compared with actual *active_power* values.

B. Time-Series Forecasting Results

The conventional time-series models such as ARIMA, ARIMAX, and SARIMAX were evaluated using the same training, validation, and test partitions. These models rely on

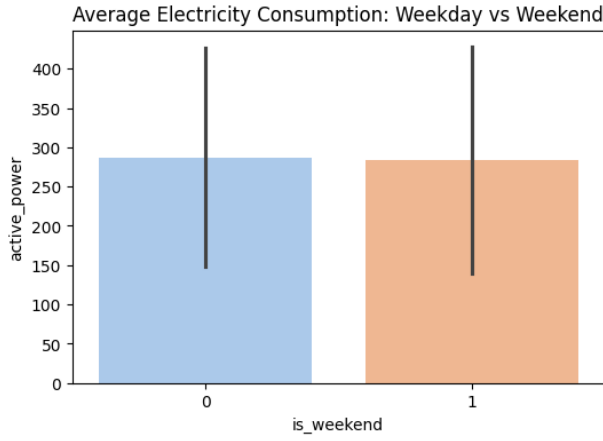


Fig. 5. Comparison of average electricity consumption between weekdays and weekends.

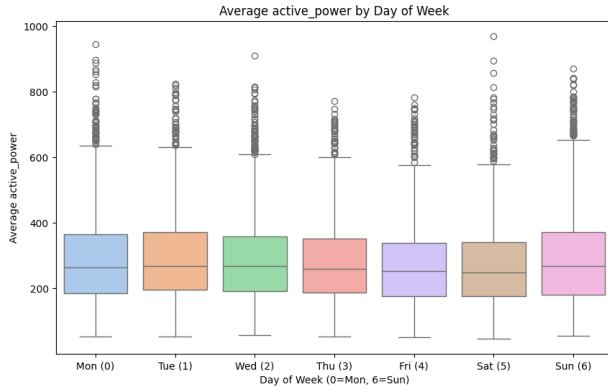


Fig. 6. Boxplot of hourly electricity consumption across days of the week.

autoregressive structure, differencing, and seasonal patterns to capture the temporal dynamics of hourly electricity consumption.

Figure 8, Fig. 9, and Fig. 10 illustrate the forecast performance of the three models on the test set. ARIMA struggles to track rapid fluctuations, producing smoothed predictions that deviate notably from observed peaks. ARIMAX improves performance by incorporating exogenous weather variables, reducing error substantially. SARIMAX reaches the best alignment with observed patterns due to its ability to model strong 24-hour seasonality present in the data.

ARIMA demonstrates the worst performance with an MAE of 95.84 and RMSE of 110.96, indicating difficulty in capturing high-frequency variability. ARIMAX substantially reduces prediction error, achieving MAE = 56.78 and RMSE = 75.03, highlighting the importance of weather variables.

SARIMAX achieves the best results among all classical models, with MAE = 54.37, RMSE = 71.23, and MAPE = 30.23%. This reinforces the role of daily seasonal cycles in hourly load behavior and demonstrates that seasonality-aware models are crucial for accurate electricity demand forecasting.

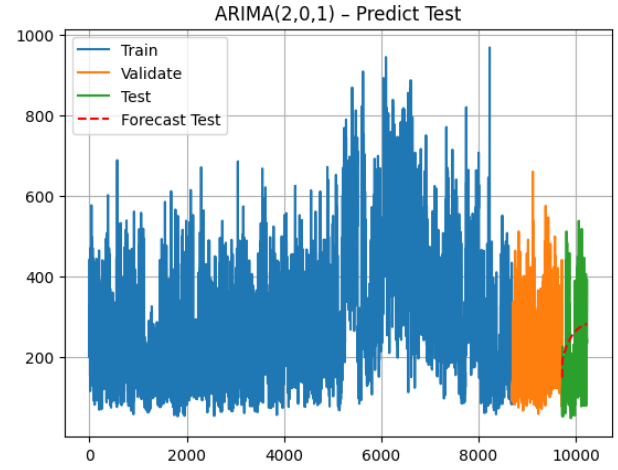


Fig. 8. ARIMA model prediction on test set.

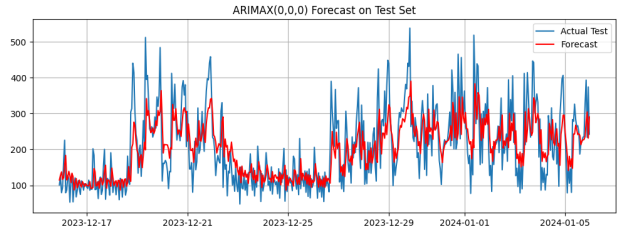


Fig. 9. ARIMAX model forecast compared to actual consumption.

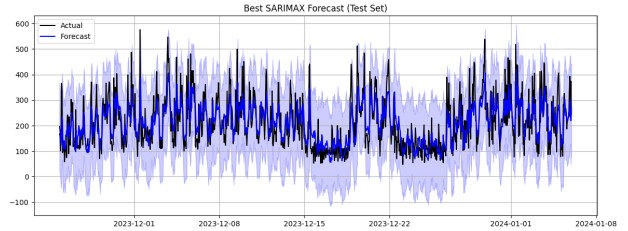


Fig. 10. SARIMAX forecast on the test set.

C. Machine Learning and Deep Learning Results

Machine learning and deep learning models were evaluated to capture nonlinear dependencies and complex temporal patterns that classical time-series models could not fully represent.

1) *Decision Tree*: The Decision Tree model provides a simple nonlinear baseline. It achieved an MAE of 11.95 and RMSE of 15.60, with an R^2 score of 0.978, reflecting moderate predictive capability but noticeable sensitivity to high-frequency fluctuations.

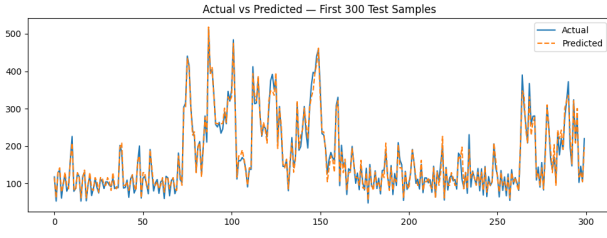


Fig. 11. Decision Tree model forecast versus actual values on the test set.

2) *Random Forest*: Random Forest achieved the strongest performance among all models, with MAE = 3.03, RMSE = 4.65, MAPE = 2.25%, and an R^2 score of 0.9980. Its ensemble structure allows it to model nonlinearities and variability more effectively.

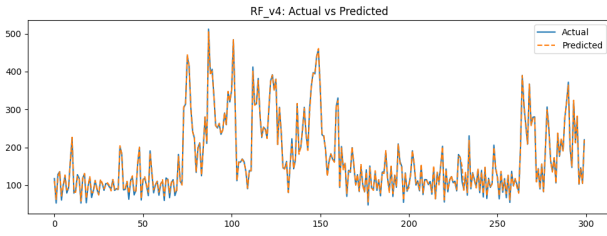


Fig. 12. Random Forest prediction performance on the test set.

3) *XGBoost*: The XGBoost model achieved MAE = 56.66, RMSE = 74.94, MAPE = 31.06%, and $R^2 = 0.3987$. While its performance is weaker than Random Forest, feature importance analysis (Fig. 13) highlights the strong influence of lag-based predictors.

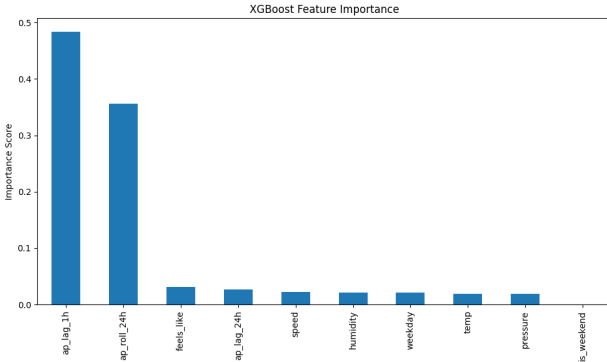


Fig. 13. XGBoost feature importance scores.

4) *LSTM Forecasting Model*: The LSTM model was evaluated to catch long-range temporal dependencies in the consumption series. The model obtained an MAE of 55.80 and an RMSE of 73.31, showing moderate performance. Although the LSTM was able to track overall consumption trends, it exhibited noticeable difficulty in predicting sharp peaks, resulting in higher error during periods of rapid load variation.

5) *Hybrid SARIMAX–LSTMX*: The hybrid SARIMAX–LSTMX model produced higher errors compared to the other

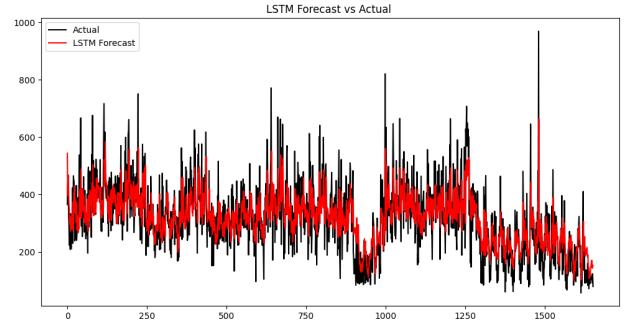


Fig. 14. LSTM forecast versus actual active_power values on the test set.

machine learning models, with MAE = 84.69, RMSE = 96.31, MAPE = 48.31%, and a negative R^2 score (−2.913). This indicates difficulty in generalizing residual patterns on the test set.

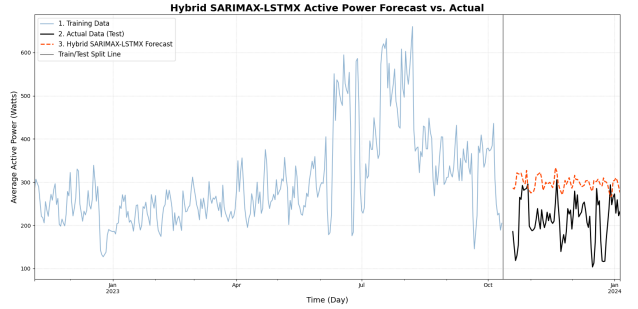


Fig. 15. Hybrid SARIMAX–LSTMX forecast versus actual values.

V. OVERALL FINDINGS AND CONCLUSION

A. Conclusion.

After implementing the research to analyze and predict household power consumption in northeastern Mexico using a 14-month dataset with one-minute recordings, integrating with many weather factors such as temperature, humidity, pressure, and speed of wind. The dataset also captures both the technical load characteristics and the meteorological conditions' impact of energy usage.

We applied both statistical models and machine learning models involving Linear Regression, ARIMA, ARIMAX, SARIMAX, Decision Trees, Random Forest, XGBoost, as well as many deep learning models like LSTM, and a Hybrid model.

As results are observed in Table IV that Random Forest is the best predicting model with MAE = 3.03, RMSE = 4.65, MAPE = 2.25%, and $R^2 = 0.9980$, which are the highest among all models. This indicates that Random Forest effectively models nonlinear relationships and interactions between energy consumption and weather factors.

It can be said that both ARIMAX and SARIMAX models also performed well, especially in capturing trends and seasonality, but their accuracy was still lower than Random Forest.

TABLE II
COMPARISON OF FORECASTING MODELS ON THE HOUSEHOLD ELECTRICITY DATASET

Metric	Statistical Models				Machine Learning / Deep Learning				Hybrid
	LinReg	ARIMA	ARIMAX	SARIMAX	DT	RF	XGB	LSTM	SARIMAX-LSTMX
MAE	60.55	95.84	56.78	54.37	11.95	3.03	56.66	55.80	84.69
RMSE	77.88	110.96	75.03	71.23	15.60	4.65	74.94	73.31	96.31
MAPE (%)	39.02	73.37	33.52	30.23	6.93	2.25	31.06	33.06	48.31
R ²	–	–	–	–	0.978	0.9980	0.3987	–	-2.913

While deep learning models like LSTM can not exceed traditional machine learning models, because the limited timeframe in 14 months and the fact that electricity load depends more on external variables than on long-term time series variables.

Generally, the results suggest that Random Forest is the most suitable model for forecasting household power usage in this dataset's context, because it gives the low errors, stability, and simplicity in implementation.

B. Limitations

Although the study achieved many positive results and provided valuable findings, there are still some limitations that need to be noted. Firstly, the biggest limitation comes from the dataset itself. Furthermore, the data was recorded from a single household in the Northeast of Mexico, so the electricity consumption reflects the characteristics and daily behavior of this household, and it is not able to be generalized to other groups of households with different sizes, structures, income levels, types of appliances used, or socio-climatic conditions. This reduces models' abstractions. Additionally, the dataset does not include important socioeconomic variables such as the number of members in the household, the level of use of electrical appliances, income, or information about holidays, vacations, or special events that may affect electricity consumption behavior. The lack of these variables makes it impossible for the model to assess the impact of macro and micro factors on electricity consumption.

Another limitation of the data is to achieve the weather information from the OpenWeather API, with the purpose of measuring errors compared to the actual climate conditions at the household location. Therefore, it can affect the quality of the forecast, especially for models that are sensitive to weather variables. Moreover, the conversion of data from minute to hourly frequency, although necessary to reduce noise and stabilize the time series, has lost some short-term features, especially sudden fluctuations that occur in very short periods that the model can no longer capture.

In addition, the analysis and modeling process also have certain limitations. Moreover, the research mainly focuses on a group of statistical and machine learning models; Other advanced techniques such as transformer models for time series and attention mechanisms have not been fully exploited. At the same time, the feature selection process has only stopped at the level of experience and basic statistical analysis, without implementing system optimization methods or

sensitivity analysis to test the influence of each variable. Also, the model has not assessed the impact of seasonal changes in data structure or socio-climate shocks, which are important factors in energy economic analysis. Finally, the study has not conducted a detailed comparison of the stability of the model over different time periods or a strong test of stationarity and time series structure for all variables.

Consequently, these limitations show that the research results need to be viewed within the scope of the data and methods used, also open up many directions for improvement for future research.

C. Insights from the study

From the process of analyzing data and comparing with predictive models, some important findings were identified. First of all, the level of electricity consumption had a significant relation with weather factors, including temperature, humidity, pressure, and speed of wind. When weather variables were included in the model, the accuracy of predictions obviously increased, particularly the error of MAPE decreasing from 73% in ARIMA to 33% in ARIMAX and 30% in SARIMAX. This showed that the condition of temperature was a significant factor in the behavior of electricity consumption, particularly on hot days when the demand for electricity increased. Secondly, the data by hour showed the cycle of electricity consumption, with high electricity consumption in noon and in the evening, and the difference between weekdays and weekends. Those characteristics reflected the habit of using electricity in a household and showed that the time variables in the day were a crucial factor in prediction. Lastly, resampling data from minutes to hours helped stand out the key relation between weather, habit, and the level of consumption, improving the efficiency of the model.

D. Recommendations

Based on the findings above, some practical suggestions could be recommended to optimize the management and prediction of electricity. With a regulatory agency and policy planner, identifying the significant effects of weather and time on the level of electricity consumption, allowing for creating policy to coordinate the electricity supply by season, particularly supplying more on hot days or periods that require high demand for electricity. Additionally, the result of the research also suggested the capability of applying the electrical price by hour (Time-of-Use Pricing), where the price will be higher

in the peak hour or lower in the midnight or the hour when the electricity demand is low. This helps to regulate the behavior of using electricity and decrease the pressure on the electrical system. With the electrical company, a Random Forest model with outstanding efficiency could be implemented in a short-term prediction system by hour to enhance the regulatory capabilities and balance the load. Moreover, the integration of predictive models in the electricity smart grid could help to find the abnormal detection, optimize the distribution, and enhance the sustainability of the electrical system. In the household scope, users could utilize the predictive model to monitor the level of electricity consumption, detect unusual situations, and regulate the habit of consumption to save electricity.

E. Future Work

In the future, the research study could be extended in some directions to increase the accuracy of prediction and enhance the capability of implementation in the electricity economy. First of all, the expansion of data scope in many households or many geographical areas helps to assess more accurately the difference in the behavior of electricity consumption and improve the generalization of models. Besides that, applying economic or social variables such as income, the members of each family, and information about holidays or events helps the models to reflect deeper on the reasons for electricity consumption from the economic outlook. Moreover, future research could be extended by implementing advanced predictive models, particularly those models based on the transformer structure for time series. Those model was proven to have the capability of catching the long-term relation and complex structure in data better than conventional models such as LSTM ,bringing higher efficiency in prediction with multiple variables. Another important improvement is implementing multi-step forecasting for a time span of 6 hours, half a of day, or a day, supporting the planning and regulatory electricity in reality. In addition, integrating an anomaly detection mechanism in the model helps to detect problems sooner, such as unusual consumption, damaged or leaking equipment.

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